MECHAEAGLES

POST-RUN DATA ANALYSIS REPORT · 2026-03-31 19:50

RUN OVERVIEW

Run	Samples	Duration	Max Speed	Max RPM	Max Temp
Run 1	118	5850 ms	32.3 mph	6183	237.2 °F

Run	Samples	Duration	Max Speed	Max RPM	Max Temp
Run 2	181	18000 ms	50.2 mph	3791	211.0 °F

CVT ENGAGEMENT

CVT-1 Time	CVT-1 RPM	CVT-2 Time	CVT-2 RPM
400 ms	3210	800 ms	2780

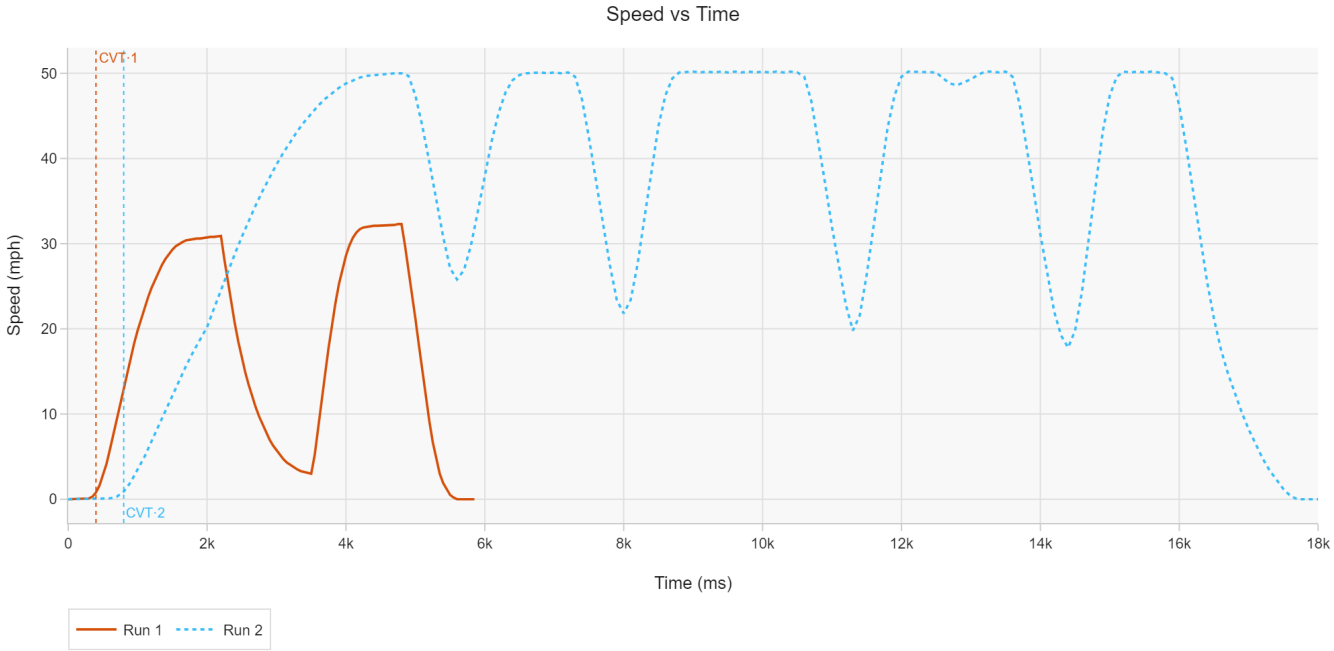
FLAGGED ANOMALIES

Run	Time (ms)	Type	Value	Threshold
Run 1	600.0	max_g	1.11	>0.75G combined magnitude
Run 1	2200.0	max_g	1.428	>0.75G combined magnitude
Run 1	3650.0	max_g	0.911	>0.75G combined magnitude
Run 1	4900.0	max_g	2.317	>0.75G combined magnitude
Run 1	5600.0	rpm_drop	1622.7	>1000 below rolling mean
Run 2	1100.0	max_g	0.751	>0.75G combined magnitude
Run 2	5100.0	max_g	0.826	>0.75G combined magnitude
Run 2	7500.0	max_g	0.87	>0.75G combined magnitude
Run 2	10800.0	max_g	0.881	>0.75G combined magnitude
Run 2	12800.0	rpm_drop	1287.0	>1000 below rolling mean
Run 2	13800.0	max_g	0.85	>0.75G combined magnitude
Run 2	16100.0	max_g	0.829	>0.75G combined magnitude

TELEMETRY PLOTS

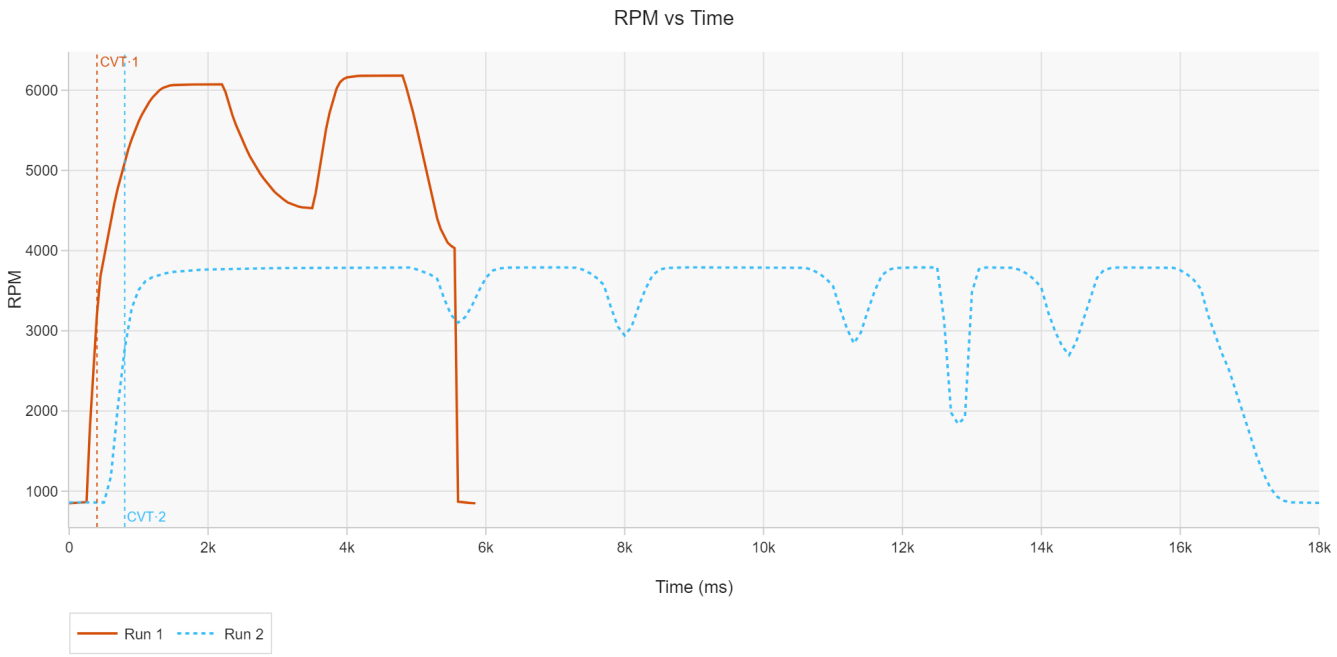
SPEED

Vehicle speed over run duration. CVT engagement point marked as dashed vertical line.



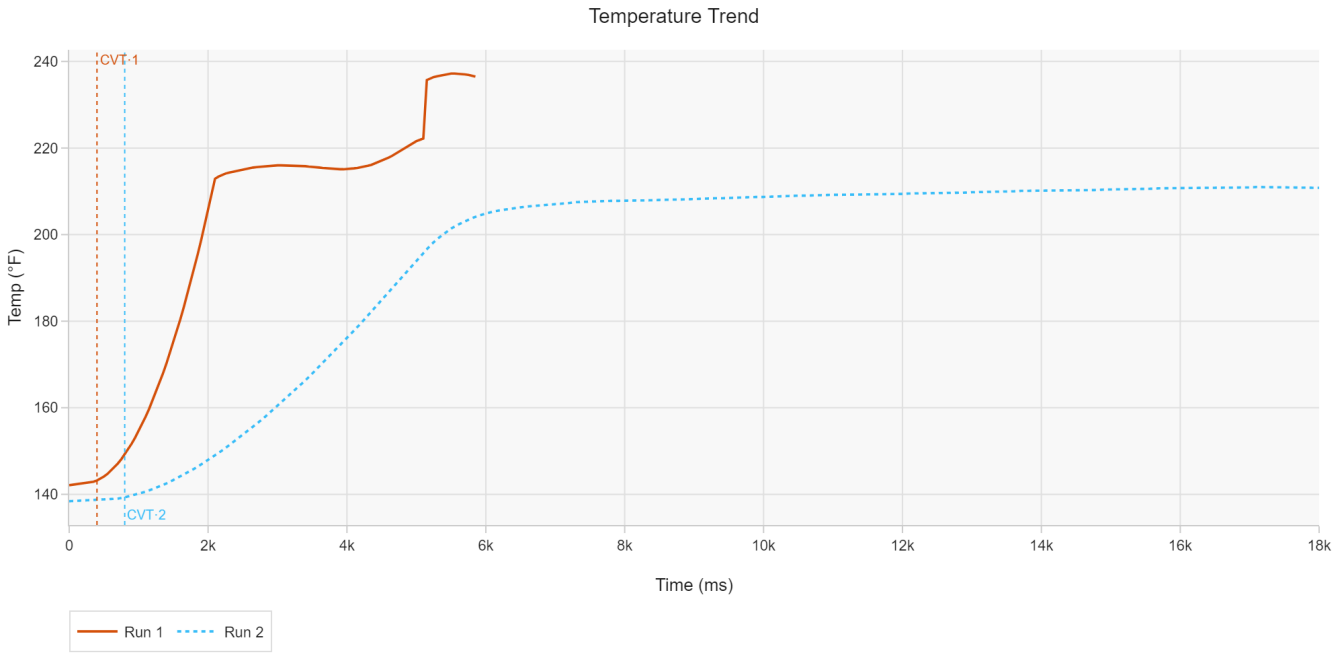
RPM

Engine RPM over run duration. Plateau region indicates belt engagement threshold.



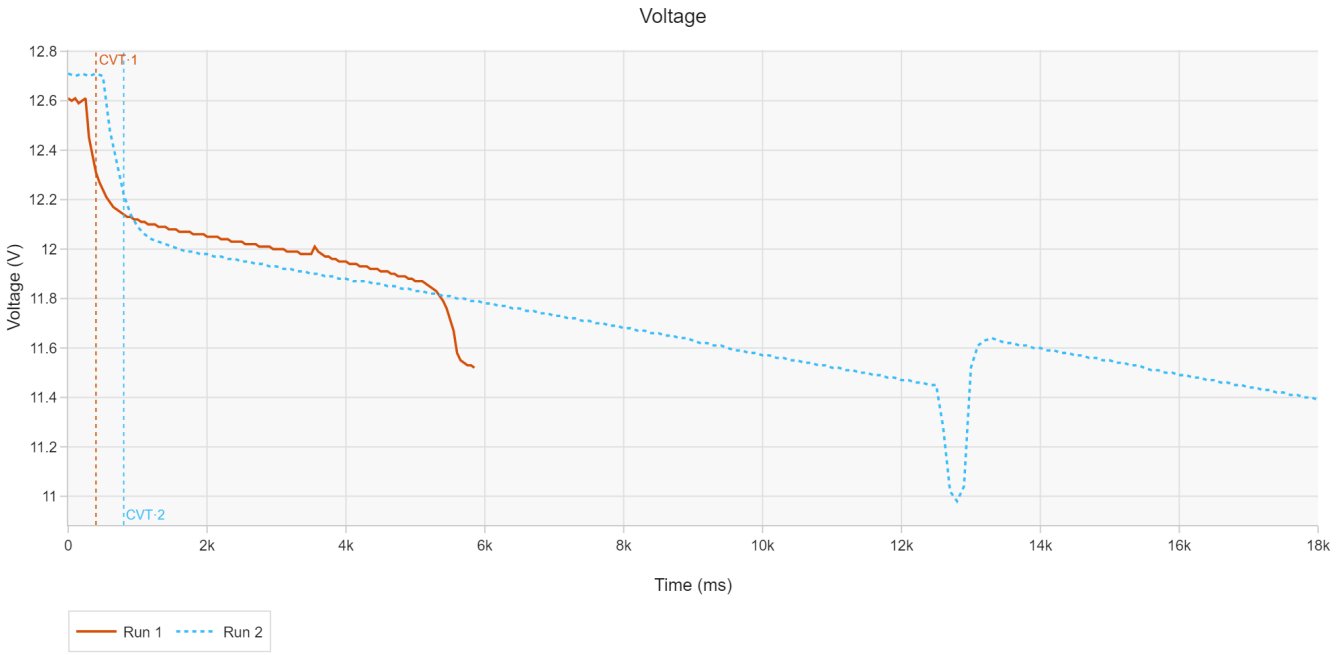
TEMPERATURE

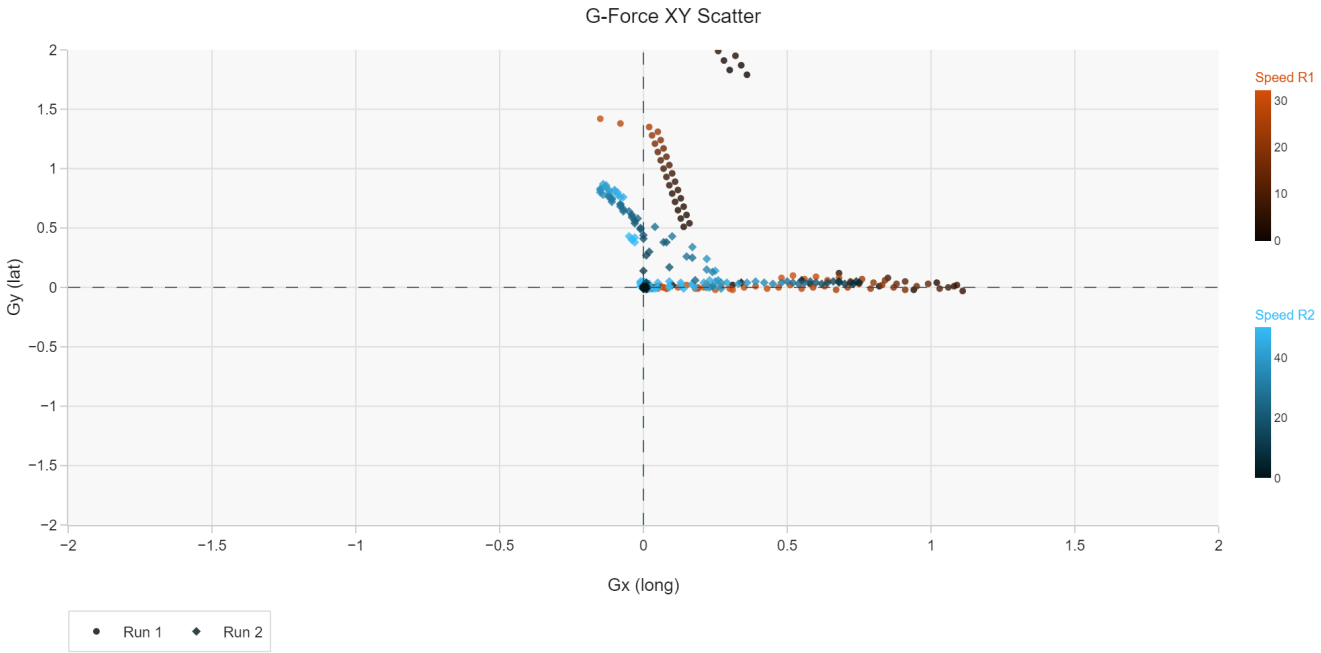
Drivetrain temperature trend across the run. Sudden rises are flagged as anomalies.



VOLTAGE

System voltage over run duration. Sustained sag may indicate electrical load issues.



G-FORCE

Lateral (Gy) vs longitudinal (Gx) G-force. Colour-coded by speed. Circle = Run 1, Diamond = Run 2.